MIL-S-55694A(EL)

26 July 1973
Superseding
MIL-S-55694(EL)
31 December 1969

MILITARY SPECIFICATION
SHELTER, ELECTRICAL EQUIPMENT
S-391( )/TGC-30

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the air or transportable facility part of the Central Office, Teletypewriter AN/TGC-30( ) which is capable of operation over one full duplex or two half duplex VF Syncronous Secure Circuits. (See 6.1 and 6.5)

1.2 Description.- Shelter, Electrical Equipment S-391( )/TGC-30 hereinafter called the shelter facility, includes a modified S-250( )/G shelter and all racks, brackets, shelves, tie-downs, electrical components and wiring to enable installation and operation of the assemblage components of the AN/TGC-30( ). (See 6.1 and 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to this document to the extent specified herein:

FSC 5410

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)
MIL-W-45205	Welding, Gas, Metal-Arc & Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable, for Structures, Excluding Armor
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys and Titanium, Spot and Seam
MIL-T-7928	Terminals, Lug and Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-S-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging and Packing of

STANDARDS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-1235	Single and Multilevel continuous sampling procedures and tables for inspection by attributes.
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

DRAWINGS

ELECTRONIC COMMAND

DL-SC-B-595491	Shelter Electrical Equipment S-391( )/TGC-30

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol shall be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article.- When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.-

3.2.1 Drawings.- The shelter facility shall be constructed in accordance with the drawings as listed on DL-SC-B-595491.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.3 Electrical Installation.-

3.3.1 Power System and Components Installation.- Installation and wiring of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly system and wiring components constituting a three wire electrical power distribution system) shall be in accordance with the drawings unless otherwise stated. The completed in-place power distribution system shall meet the following requirements: (See 4.7.1)

3.3.1.1 Continuity of Conductors.- There shall be point-to-point circuit continuity on all power-line circuits and the resistance of the continuous circuit shall not exceed 3 ohms when tested in accordance with 4.7.1 and 4.7.1.1.

3.3.1.2 Dielectric Breakdown Voltage.- The dielectric strength of insulation of all power wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds when tested in accordance with 4.7.1 and 4.7.1.2.

3.3.1.3 Insulation Resistance.- The resistance of insulation between any two of three circuit conductors and also between each of the two ungrounded power-line conductors and shelter facility ground terminal shall not be less than 100 megohms when tested in accordance with 4.7.1 and 4.7.1.3.

3.3.2 Power Distribution and Operational Functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as denoted by power system schematic plate when tested in accordance with 4.7.2.

3.3.3 Signal (audio) System Wiring and Components Installation.- Installation and wiring of signal wires, cables and wiring components (including binding posts, connectors, plugs, jacks, receptacles, switches, terminal strips, terminal blocks, and wiring devices) shall be in accordance with the drawings unless otherwise stated. The completed in-place signal system installation shall meet the following requirements: (See 4.7.3)

3.3.3.1 Continuity of Conductors.- There shall be circuit continuity between both end terminations of all signal-line conductors and the resistance of the continuous circuit shall not exceed 5 ohms when tested in accordance with 4.7.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.3.3.2 Dielectric Breakdown Voltage.- The dielectric strength of insulation of all signal wires, cables and components, shall withstand a potential of 500 volts DC applied continuously for not less than 5 seconds when tested in accordance with 4.7.3.2.

3.3.3.3 Insulation Resistance.- The resistance of insulation between each signal-line conductor and all other signal-line conductors connected to ground shall not be less than 100 megohms when tested in accordance with 4.7.3.3.

3.3.4 Cable Assemblies, Power and Audio Frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings unless otherwise stated and shall meet the continuity, dielectric, insulation resistance and pull-out requirements of MIL-C-3885. (See 4.4)

3.3.5 Suppression of Electromagnetic Interference.- The motors in the ventilating equipment and heaters used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings. (See 4.4).

3.4 System Operation.- With all operational equipments of the AN/TGC-30 assemblage installed, the operational requirements of all power and audio cable assemblies, switches, connectors, and patch panels, shall be met when tested in accordance with 4.8.

3.5 Vehicular Transportation.- The shelter facility shall be capable of being transported over cross-country terrain without any extra tieing, taping, etc. by military vehicles without sustaining any damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.9)

3.6 Rail Transportation.- The shelter facility without any extra tieing, taping, etc., shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail and shall be subjected to and withstand the test of paragraph 4.10 without sustaining any buckling, cracking, delamination, or other permanent deformation.

3.7 Drops, Flat and Rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto hard concrete. After all drop tests of 4.11.1 and 4.11.2 there shall be no buckling, cracking, delamination or other permanent deformation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.12. During and after the test there shall be no leakage into the facility or its walls, floor, roof, or doors.

3.9 Fording.- The shelter facility shall be capable of withstanding the test of 4.13. During and after the test there shall be no leakage into the facility or its walls, floor, roof or doors.

3.10 Mounting Inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and, when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.14. There shall be no damage or permanent distortion to the shelter mounting surface or the insert threads or head and the inserts shall not have rotated from its original position. Insert threads shall be free of paint.

3.11 Cleaning.- (See 3.22)

3.11.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.11.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.12 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.13 Marking.- (See 3.22 and 4.4)

3.13.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.13.2 Nameplates.- Shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft Loading Data Plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be determined by the contractor using the first article sample.

3.13.4 Interior and Exterior Marking.- All interior and exterior markings shall be as defined on the drawings.

3.14 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. (See 3.22)

3.15 Seal, Water Vapor.- All joints of the framing for the exhaust blowers, power and signal entrance boxes, and any other framed openings shall be sealed to provide a barrier against the entrance of water vapor to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be a one-part sealant, Silicone Construction Sealant SE-1204, as made by General Electric Corporation, or 732 as made by Dow Corning or equal using the manufacturers recommended primer. (See 3.22)

3.16 Securing of Parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests. (See 3.22)

3.16.1 Screws, Threaded Devices, and Related Parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.17 Soldering.- (See 3.22)

3.17.1 Solder and Flux.- Solder and flux shall be as defined on the drawings.

3.17.2 Acid or Acid Salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over-heated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.18 Tropicalization of Material.- Canvas , webbing, wood, etc., shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows: (See 3.22)

3.18.1 Treating Materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of Treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposed to use. (See 6.2e)

3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Inert gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL. B and MIL-W-45205 CL. B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusion, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 3.22 and 4.4)

3.20 Wiring and Cabling.- Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy. (See 3.22)

3.20.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided.

a. To prevent undue stress on cable forms, wires and connections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

b. To permit removal from rack of operational equipments, having wiring or cables to connectors at back of equipment, so that cables may be connected or disconnected.

c. To facilitate field repair of broken or cut wires.

d. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.20.4 Crimping of Solderless Terminal Lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.4 and 4.15)

3.20.5 Identification of Wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

3.21 Interchangeability.- Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.16) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs: (See 4.17)

Source: https://assist.dla.mil -- Downloaded:-202610-26T04:18Z
Check the source to verify that this is the current version before use.

3.2 Construction
3.3 Electrical installation
3.10 Mounting inserts
3.11 Cleaning
3.12 Finish
3.13 Marking
3.14 Riveting
3.15 Seal, water vapor
3.16 Securing of parts
3.17 Soldering
3.18 Tropicalization
3.19 Welding
3.20 Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Classification of Inspection.- Inspection shall be classified as follows:

a. First article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.18)

4.3 First Article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.3.1 First Article Units.- The contractor shall furnish one (1) first article unit of the complete S-391( )/TGC-30.

4.3.2 First Article Inspection.- The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First Article Data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I First Article Inspection

Inspections	Rqr Para	Test Para
1. Inspection covered by subsidiary documents	-	4.4
2. Interchangeability	3.21	4.16
3. Mounting Insert	3.10	4.14.1
4. Solderless-terminal-lug connectors	3.20.4	4.15
5. Visual and Mechanical	3.22	4.17
6. Power systems and components installation	3.3.1	4.7.1
7. Power distribution and operational function	3.3.2	4.7.2
8. Signal (audio) system wiring and components installation	3.3.3	4.7.3
9. System operation	3.4	4.8
10. Vehicular transportation	3.5	4.9
11. Rail transportation	3.6	4.10
12. Flat drop	3.7	4.11.1
13. Rotational drop	3.7	4.11.2
14. Watertightness	3.8	4.12
15. Fording	3.9	4.13
16. Repeat 6. Power systems and components installation	3.3.1	4.7.1
17. Repeat 7. Power distributional and operational function	3.3.2	4.7.2
18. Repeat 8. Signal (audio) system wiring & components installation	3.3.3	4.7.3
19. Repeat 9. System operation	3.4	4.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.4 Inspection Covered by Subsidiary Documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where Required
Cable assemblies	3.3.4
Suppression, EMI	3.3.5
Finish	3.12
Marking	3.13
Welding	3.19
Terminals	3.20.4

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

4.5.1.1 Order of Inspection Within Group A.- Group A inspection shall be performed in the order listed in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

TABLE II - Group A Inspection

Inspection	Rqr Para	Insp Para	AQL % Def Major	Minor
Visual and Mechanical	3.22	4.17	1.0	4.0
Electrical installation				**
Power systems and component installation				
Continuity of conductors	3.3.1.1	4.7.1.1		
Dielectric breakdown voltage	3.3.1.2	4.7.1.2		
Insulation resistance	3.3.1.3	4.7.1.3		
Power distribution and operational function	3.3.2	4.7.2		
Signal (audio) system wiring and components installation				
Continuity of conductors	3.3.3.1	4.7.3.1		
Dielectric breakdown voltage	3.3.3.2	4.7.3.2		
Insulation resistance	3.3.3.3	4.7.3.3		

1.0 for the entire group of power and signal wiring.

** All defects are major.

4.5.2 Group B Inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.5.2.1 Group B Sampling Plans.- The Group B sampling plans for the AQL's listed in Table III of this specification shall be as follows:

AQL	Inspection Level
4.0%	S-4
2.5%	S-4

4.5.2.2 Order of Inspection Within Group B.- Group B inspection shall be performed on the same equipment in the order listed in Table III.

TABLE III - Group B Inspection

Inspection	Rqr Para	Insp Para	AQL % Def (See 4.5.2.1)
Mounting insert	3.10	4.14.2	4.0
Interchangeability	3.21	4.16	4.0
Watertightness	3.8	4.12	2.5
Fording	3.9	4.13	2.5

NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in 4.5.2.2.

4.5.3 Group C Inspection.- This inspection shall be as listed in Table IV.

TABLE IV - Group C Inspection

Inspection	Rqr Para	Insp Para
Solderless-terminal-lug connection	3.20.4	4.15
Construction verification	3.2	4.6
Mounting inserts	3.10	4.14.1
System operation	3.4	4.8
Vehicular transportation	3.5	4.9
Railroad transportation	3.6	4.10
Drops, Flat and Rotational	3.7	4.11

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3. The order of inspection within Group C for all inspections listed shall be as listed in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

Units subjected to Group C Drop Tests shall subsequently be subjected to and meet the watertightness and fording tests of 4.12 and 4.13.

4.5.3.1 Sampling for Inspection of Solderless-Terminal-Lug Connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.2 Sampling for Inspection of Construction Verification.- The Equipment Verification Review, in accordance with paragraph 4.6 shall be performed once on each part and assembly, randomly selected, and on their installation into one of first 10 shelter facilities produced. This inspection shall not be performed on the preproduction unit. This inspection shall be performed as the selected shelter is being assembled.

4.5.3.3 Sampling for Inspection of Drop Tests.- One sample, representative of the first 10 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the truck and rail transportation tests of 4.5.3.4. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.4 Sampling for Other Inspection.- For inspection of tests listed below, one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the Government.

a. Mounting inserts
b. System operation
c. Truck transportability
d. Railroad transportability

4.5.3.5 Group C Failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.5.4 Reinspection of Conforming Group C Sample Units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.5.5 Alternate Group A Audit and Group B Inspection Procedures.- The Contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plans from MIL-STD-105. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes". Sampling plans for the AQLs listed in Table II Group A inspection and Table III Group B inspection of this specification shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=1/2. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production Contractor shall notify the contracting officer in writing which method he intends to use. The method may not be changed during production without written approval of the Government.

4.6 Construction Verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (see 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). EVR shall consist of the following:

a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in sub-assemblies, and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

b. An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, component, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6 a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the Government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to Government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts, components of shelter facilities, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

4.7 Electrical Installation Tests.- The tests of 4.7.1 through 4.7.2.3 shall be performed on each shelter facility prior to installation of other major components comprising the AN/TGC-30. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to the shelter facility shall be 115 VAC, 60 cycles, single phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.7.1 Power System and Components Installation Test.- Prior to conducting power wiring and components tests of 4.7.1.1, 4.7.1.2 and 4.7.1.3, the following shall be accomplished.

a. Remove or disconnect the following electrical loads:

(1) Fluorescent lamps and lamp starters.

(2) Incandescent cold start lamps

(3) Power distribution box neon indicator lamps. (Red & Blk)

(4) Power indicator neon lamp

(5) Blower

(6) Heater

(7) Air conditioner

b. Remove black power distribution box cover panel and disconnect both conductors from the voltmeter terminals and step down transformers. Position or insulate leads so that they do not short to each other.

c. Remove red power distribution box cover panel and disconnect both conductors from the voltmeter terminals and frequency meter terminals. Position or insulate leads so they do not short to each other.

4.7.1.1 Continuity of Conductors.- Continuity measurements which satisfy the requirements 3.3.1.1 shall be made on each power-line circuit as indicated by the following paragraphs using an aural or visual indicating continuity test instrument and suitable test jig. The instrument shall produce a test current between 0.2 and 2.0 amperes AC or DC when circuit continuity exists.

a. Remove and extend all connected power receptacles from wireway so that the outlet mounting brackets do not make electrical contact to the wireway. (See Figure 1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

b. Place the following circuit controls in the ON position.

(1) Main circuit breakers (blk & Red Power Dist.)

(2) Branch circuit breakers

(3) Fluorescent lamp switch

(4) Incandescent cold start switch

(5) Bypass/Blackout switch

(6) Blower switch

(7) Night light switch

(8) Place switch in Red power distribution panel to equipment position.

c. Measure continuity from pin 1 (of power IN connector in power entrance box) to the following: brass contact on each receptacle; black wire termination at each fluorescent lamp; and black wire termination at each incandescent lamp.

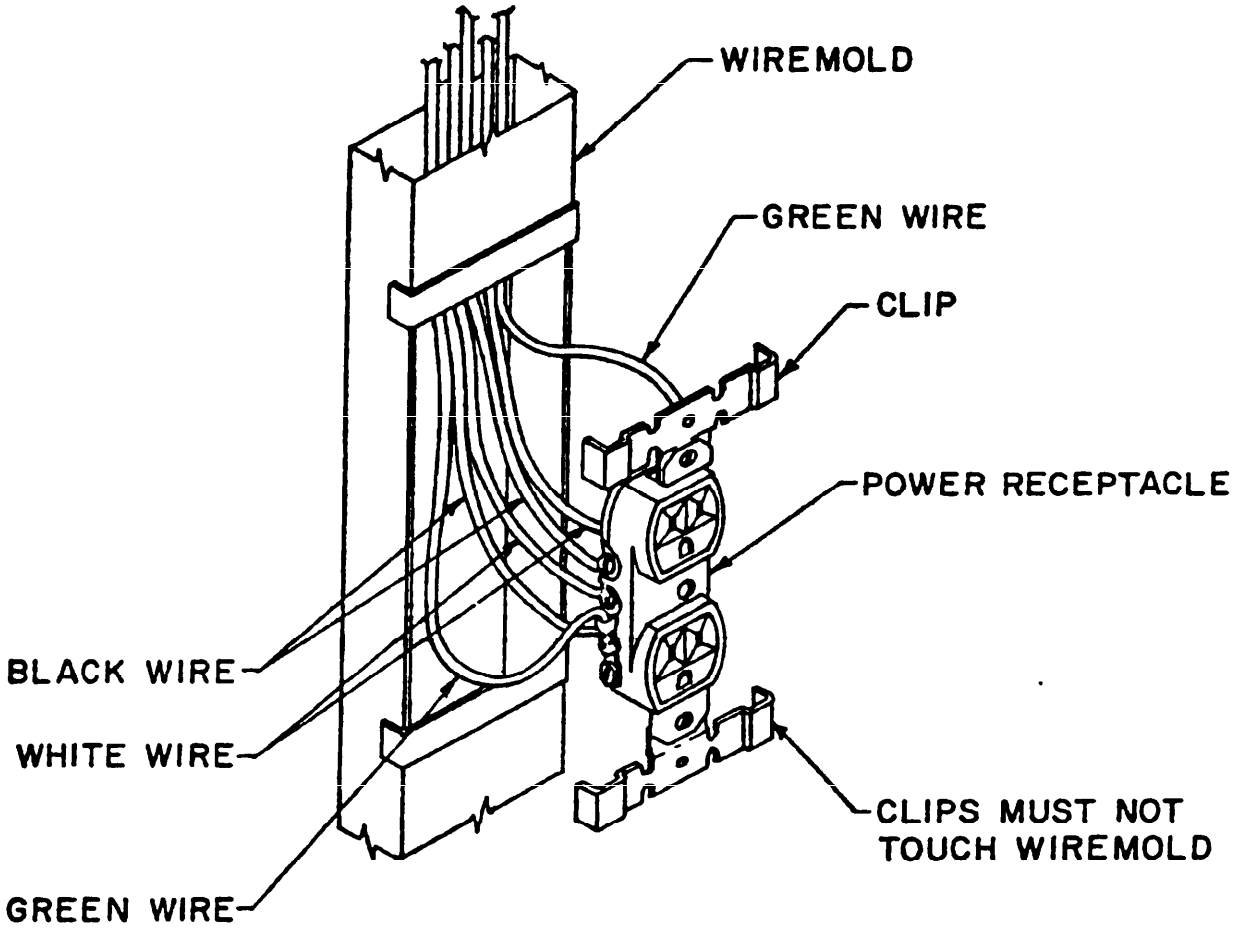
d. Place all controls listed under procedure 4.7.1.1b in OFF position, except main circuit breakers. (Red & Blk Distribution Panel)

e. Measure continuity from pin 2 (of power IN connector in Power Entrance Box) to the following: silver contacts on each receptacle; white wire termination at each fluorescent lamp and white wire termination at each incandescent lamp.

f. To check continuity to heater receptacle place switch in red power distribution to HEATER position and repeat procedure 4.7.1.1c and e.

g. Place main circuit breakers (Red & Blk Distribution Panel) in OFF position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



POWER RECEPTACLE REMOVED
& EXTENDED FROM WIREMOLD

FIG. 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

h. Measure continuity from pin 4 (ground) (of power IN connector) to each ground terminal on receptacles.

i. Measure continuity from pin 4 (ground) (of power IN connector) to shelter ground terminal located in power entrance box panel.

j. Measure continuity from pin 1 of power IN connector to pin 1 of power OUT connector; from pin 2 of power IN connector to pin 2 of power OUT connector; and from pin 4 of power IN connector to pin 4 of power OUT connector.

k. Replace and secure power receptacle removed by procedure 4.6.1.1a.

l. Measurements made by 4.7.1.1c, e, h, i, j shall indicate existence of electrical circuit continuity. (See 3.4.1.1).

4.7.1.2 Dielectric Breakdown Voltage.- Dielectric strength of insulation tests which satisfy the requirements of 3.3.1.2 shall be made on each powerline circuit indicated by the following paragraphs (Method 301 of MIL-STD-202, shall be employed using a nondestructive DC dielectric tester):

a. Place all circuit controls listed in procedure 4.7.1.1b in the ON position except fluorescent lamp switch.

b. Connect tester between Pins 1 and 4 of power IN connector J1 in power entrance box. Adjust tester for a value of 600 volts DC and maintain this potential for a period of 30 seconds. The requirements of 3.3.1.2 shall be met.

c. Place fluorescent lamp switch to the ON position.

d. Repeat procedure 4.7.1.2b with tester connected between Pins 1 and 2 of power IN connector.

e. Repeat procedure 4.7.1.2b with tester connected between Pins 2 and 4 of power IN connector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.7.1.3 Insulation resistance.- Insulation resistance tests which satisfy the requirements of 3.3.1.3, shall be made on each power-line circuit indicated by the following paragraph (Method 302 of MIL-STD-202 shall be employed using a megohmmeter or equivalent tester):

a. Connect tester between pin 1 and pin 2 of power IN connector in power entrance box. Adjust tester for a value potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.1.3 shall be met.

b. Repeat procedure 4.7.1.3a with tester connected between pin 1 and pin 4 of power IN connector.

c. Repeat procedure 4.7.1.3a with tester connected between pin 2 and 4 of power IN connector.

NOTE: To eliminate danger of shock.- Discharge conductors following each test.

4.7.2 Power distribution and operational functions tests which satisfy the requirements of 3.3.2 shall be made as indicated by the following paragraphs:

4.7.2.1 Test preparation.-

a. Replace or reconnect all electrical loads and line voltmeter and frequency meter removed or disconnected by procedure 4.7.1a, b and c.

b. Replace power distribution box covers. (Red and Blk)

c. Place circuit breakers and switches in the OFF position.

d. Reset circuit breaker in the power entrance box in the ON position.

e. Place switch in red power distribution box to equipment position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

f. Place a 115 volt AC test lamp with grounding type plug into each convenience and equipment receptacle.

g. Open (and secure open) outer exhaust port covers of blower.

h. Place by-pass blackout switch in the ON position.

4.7.2.2 Line voltmeter, frequency meter and indicator lamp.-

a. Apply 115 volt AC (nominal), 60 cycles, power to power IN connector in power entrance box. Power indicator neon lamp shall light; facility voltmeter in Blk power distribution panel shall indicate power to the shelter. Frequency meter in Red power distribution panel shall indicate 60 Cps. Test lamp in receptacle in power entrance box shall light.

b. Measure specific line voltage by connecting a calibrated external AC voltmeter between pin 1 and pin 2 of power OUT connector.

c. Read voltage indicated on facility voltmeter. The voltage read shall be within 3% of the voltage read by 4.7.2.2b.

d. Place main circuit breaker in ON position (black distribution panel).

e. Place circuit breaker in OFF position (red distribution panel).

4.7.2.3 Circuit breakers and switches.-

Circuit No. 1.- (Lights circuit breaker)

a. Place circuit breaker in the ON position. Neon lamp shall light. Push in Entry Button in Power Entrance Box. Buzzer shall sound.

b. Place fluorescent light switch in ON position. All fluorescent lamps shall light.

c. Place incandescent light switch in ON position. Incandescent lamps shall light.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.-

d. Place circuit breaker in OFF position. Neon lamp and all fluorescent and incandescent lamps shall go out. Push in entry button in power entrance box. Buzzer shall not sound.

e. Place circuit breaker in ON position. With shelter door open, place by-pass blackout switch in OFF position. All fluorescent and incandescent lamps shall go out.

f. Close shelter door. All fluorescent and incandescent lamps shall light.

g. Open door and place by-pass blackout switch in ON position.

Circuit No. 2.- (Air conditioner circuit breaker)

a. Place circuit breaker in ON position. Neon lamp and test lamp shall light.

b. Place circuit breaker in OFF position. Neon lamp and test lamp shall go out.

Circuit No. 3.- (TH-22 circuit breaker)

a. Place circuit breaker in the ON position. Neon lamp shall light and test lamp in each receptable shall light.

b. Place circuit breaker in the OFF position. Neon lamp shall go out and test lamp in each receptacle shall go out.

Circuit No. 4.- (CV-425 circuit breaker)

a. Place circuit breaker in the ON position. Neon lamp shall light and test lamp in each receptacle shall light.

b. Place circuit breaker in the OFF position. Neon lamp shall go out and test lamp in each receptacle shall go out.

Circuit No. 5.- (Blower circuit breaker)

a. Place circuit breaker in ON position. Neon lamp shall light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

b. Place blower switch in ON position. Blower shall operate and exhaust port cover shall open.

c. Place circuit breaker in OFF position. Neon lamp shall go out and blower shall stop and exhaust port cover shall close.

Circuit No. 6.- (Convenience Check and intercom circuit breaker)

a. Place circuit breaker in ON position. Neon lamp shall light and test lamp in each receptacle shall light.

b. Place circuit breaker in OFF position. Neon lamp shall go out. Test lamp in each receptacle shall go out.

Circuit No. 7.- (Main circuit breaker) (black distribution panel)

a. Place all circuit breakers (except in red distribution panel) in ON position. All lamps shall light. Blower and air conditioner shall operate. Test lamp in each receptacle (except those on red power) shall light.

b. Place circuit breaker in OFF position. All lamps shall go out. Blower and air conditioner shall cease to operate. Test lamp in each receptacle shall go out.

Circuit No. 8.- (Circuit breaker in red distribution box)

a. Place circuit breaker in ON position. Neon lamp for equipment shall light and test lamp in each receptacle (on red power) shall light.

b. Place switch in HEATER position. Neon lamp for heater receptacle and test lamp shall light.

c. Place circuit breaker in the OFF position. Neon lamp for heater receptacle and test lamp shall go out. Neon lamp for equipment shall not light and test lamp in each receptacle (on red power) shall not light.

Source: https://assist.dla.mil -- Downloaded: 20261026T04:18Z
Check the source to verify that this is the current version before use.

d. Disconnect all test equipment and secure operational test.

4.7.3 Signal (Audio) system wiring and components installation tests.- Prior to conducting signal wiring and components tests of 4.7.3.1, 4.7.3.2 and 4.7.3.3 the following shall be accomplished:

a. Provide terminal boards, for terminating all circuits as shown on signal schematic diagram furnished with model in order to accomplish testing.

b. Terminal boards or equivalent shall be provided for terminating circuits of equipment not part of facility as follows:

(1) FIG 2 Switchboard, Telephone SB-22A/PT

(2) FIG 3 Converter-Telegraph Telephone Signal CV-425/U.

(3) FIG 4 Terminal, Telegraph TH-22/TG

c. Plug in cables from patch panel TH-22 loop side to SEND/REC jacks position 1 & 2 of terminal block (TH-22) FIG 4.

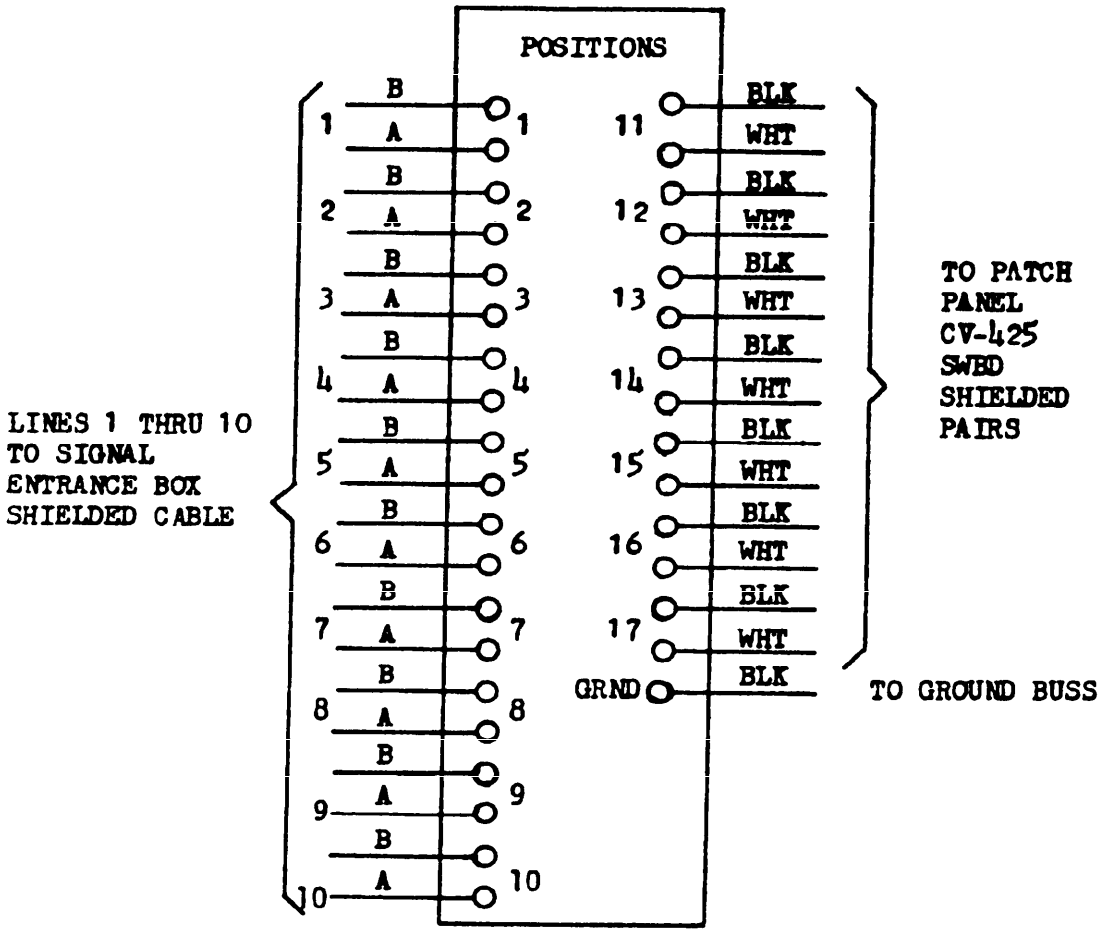
4.7.3.1 Continuity or conductors.- Continuity resistance which satisfies the requirements 3.3.3.1 shall be made on each signal circuit indicated by the following paragraphs (using a low resistance continuity tester or equivalent, producing a test current of not more than .010 amps).

a. At interconnecting box, position #1 from binding post E1 to connector J8, pin C; E2 to J7, pin B; E3 to pin A of J3, J4, J7 and J8, E4 to J3, pin B; E5 to J4, pin C; J7 pin C to J8, pin B; J4, pin B to J3, pin C.

b. Repeat a above for interconnecting box, position #2.

c. At switch box, position #1, from J1, pin B to J3, pin B; J1, pin C to J2, pin B, J2, pin A to J3 pin A. With switch in the ON position using ohmmeter measure 27,000 ohms ±5% from J1, pin B to J3 pin B.

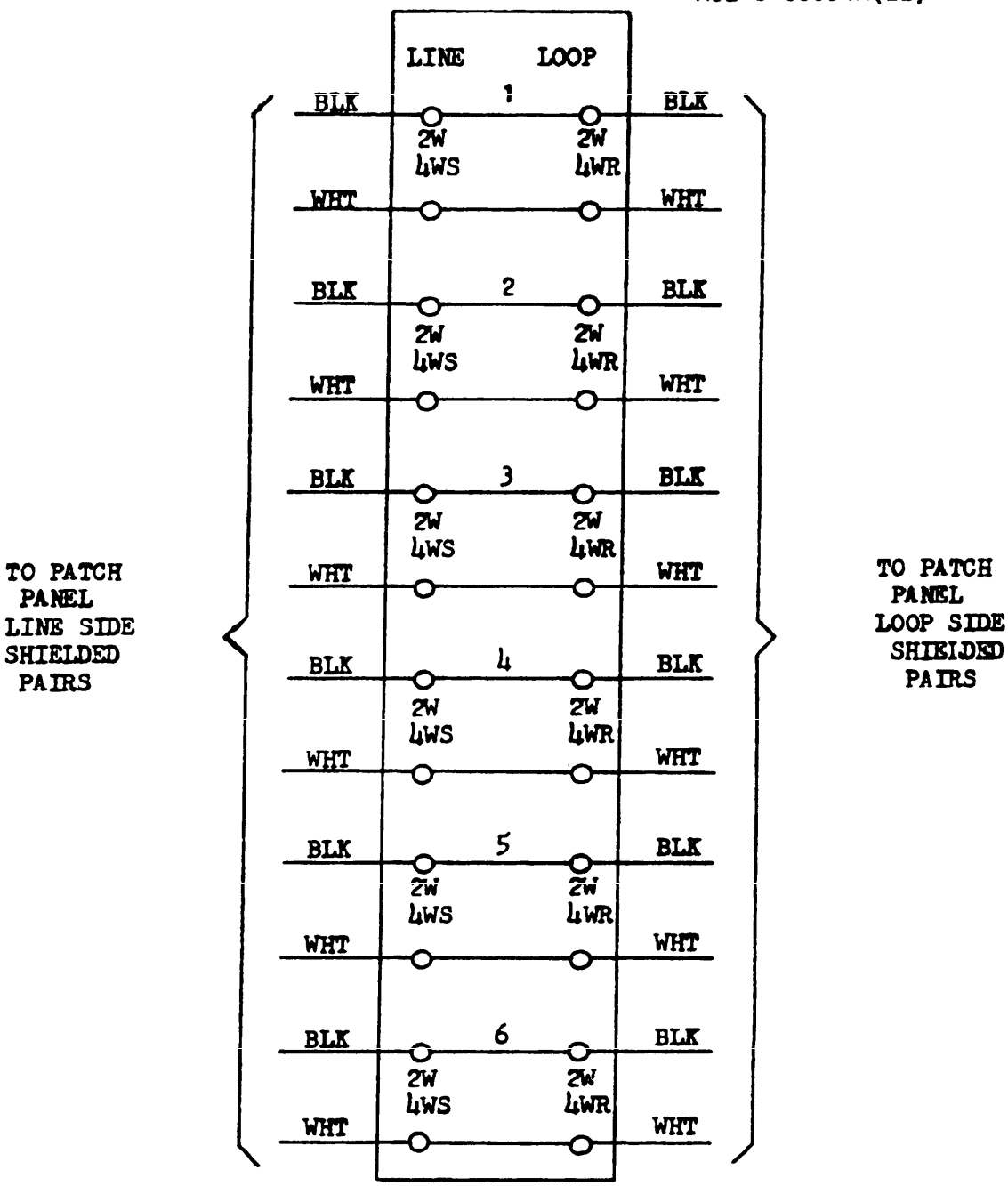
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



TERMINAL BOARD (SB-22()/PT CIRCUITS)

FIG.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



TERMINAL BOARD (CV-425/U CIRCUITS)
FIG.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

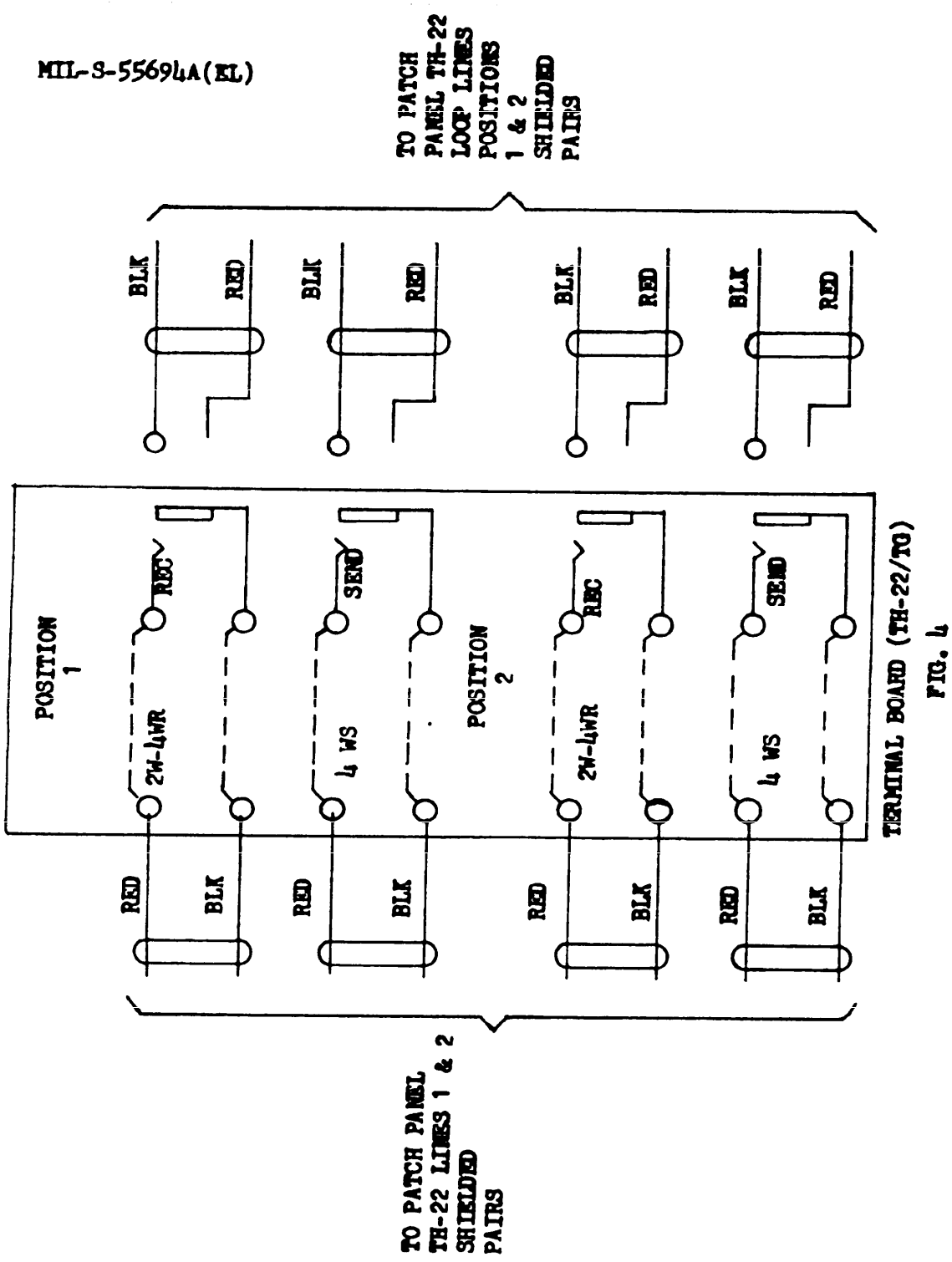


FIG. 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z.
Check the source to verify that this is the current version before use.

d. Repeat c above for switchbox, position #2.

e. At intercom and telephone box, place switch in intercom position. Short wires for TA-312, at binding post U-106( )/U check continuity between conductors 25A and 25B.

f. Place switch in telephone position, short wires for LS-147( )/U, at binding post U-106/U check continuity between 26A and 26B.

g. Place switches S1 and S2 on signal entrance box to cable position.

h. Lines 1 to 10 from U-187( )/U thru U-106/U to terminal block (SB-22) FIG 2 positions 1 to 10.

i. Lines 11 to 16 from U-187( )/U thru U-106/U thru patch panel lines 11 to 16 thru terminal block (CV-425( )/U) FIG 3 lines 1 to 6, thru patch panel loop side 1 to 6 terminal block (SB-22( )/PT) FIG 2 positions 11 to 16.

j. Lines 17 to 20 from U-187( )/U thru patch panel lines 17 to 20 to terminal block (TH-22( )/TG) FIG 4 Position 1 and 2. Thru SEND/REC of positions 1 and 2 thru patch panel loop side TH-22 to interconnecting boxes 1 and 2.

k. Repeat j above with switch 1 and 2 on signal entrance box to binding post position, test from U-106/U.

l. Lines 21 to 24 at binding post U-106( )/U between 21B and 21A, 22B and 22A, 23B and 23A and 24B and 24A.

m. For SB-22( )/PT opr circuit, at terminal block (SB-22( )/PT) FIG 2 test across position 17.

n. Insert plug type PJ-047 in the following jacks on Patch Panel in sequence indicated below to check break operation (open) of line circuits 11 to 20 tested in i and j above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

CV425	Line 11,	Line 1,	Loop 1,	SWBD	PR11
"	" 12	" 2	" 2	"	" 12
"	" 13	" 3	" 3	"	" 13
"	" 14	" 4	" 4	"	" 14
"	" 15	" 5	" 5	"	" 15
"	" 16	" 6	" 6	"	" 16

Pos. 1, Line 17, 2W-4WR; TH-22 Line 4WR; Pos. 1, TH-22 Loop Rec, Equip Rec.
Pos. 1, Line 18, 4WS; TH-22 Line 4WS; Pos. 1, TH-22 Loop Send, Equip Send.
Pos. 2, Line 19, 2W-4WR; TH-22 Line 2W-4WR; Pos. 2 TH-22 Loop Rec, Equip Rec.
Pos. 2, Line 20, 4WS; TH-22 Line 4WS; Pos. 2, TH-22 Loop Send, Equip Send.

o. Measure continuity from black grounding bus and red grounding bus to shelter ground termination in power entrance box.

p. Measure continuity from interconnecting box 1 & 2 plug E3 to shelter ground terminal.

q. Measure continuity from ground lead for equipment case to red or black grounding bus as applicable.

r. Measure continuity of cable shields to ground bus bar in signal entrance box.

s. Disconnect all plugs from patch panel.

4.7.3.2 Dielectric breakdown voltage.- Dielectric strength of insulation tests which satisfy the requirements of 3.3.3.2 shall be made for each signal circuit indicated by the following paragraphs (Method 301 of MIL-STD-202 shall be used utilizing a non-destructive dielectric test instrument and suitable test jigs or an automatic tester may be used subject to government approval): Prior to conducting test, insert plug type PJ-047 at Patch Panel in spare lines 21 to 24.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

a. Connect the following to facility ground terminal: Pin 1A thru 24B of signal connector J1.

b. Connect tester between facility ground terminal and Pin 1A of signal connector J1. Remove ground connection from pin 1A of signal connector J1.

c. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds the requirements of 3.3.3.2 shall be met.

d. Disconnect tester lead from pin 1A of signal connector J1 and re-established ground connection of Pin 1A of Signal connector J1.

e. Repeat procedures 4.7.2.2b thru d for each remaining (pins 1B thru 24B of signal connector J1) each time ungrounding the pin of signal connector J1 correlating with the circuit being tested and reconnecting pin to facility ground terminal following completion of test.

NOTE: To eliminate danger of shock.- Discharge conductors following each test.

4.7.3.3 Insulation resistance.- Insulation resistance which satisfy the requirements of 3.3.3.3 shall be made on each signal circuit indicated by the following paragraphs (Method 302 of MIL-STD-202 shall be used, utilizing a megohmmeter or equivalent tester, and suitable test jig or an automatic tester subject to government approval):

a. Repeat procedures 4.7.2.2a thru e except 4.7.2.2c which shall be replaced by 4.7.2.3b below.

b. Adjust tester to 500 volts DC output. The requirements of 3.3.3.3 shall be met.

NOTE: To eliminate danger of shock.- Discharge conductors following each test.

4.8 System operation test.-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.8.1 Signal Circuit Preparation Prior to Conducting test.-

a. Connect distribution box J-1077/U using 26 pair cable CX-4760 to signal 1 connector IN signal entrance box.

b. Plug in all power cords from equipment to their respective outlets, place circuit breakers and power switches associated with the equipments to the ON position.

c. Prepare a test tape containing a message utlizing all letters.

d. At TT-76( )/GGC Positions 1 & 2 install TD send, TR send and REC wire pairs to their respective terminals, and install tape roll.

e. At TT-98( )/FG positions 1 & 2 install keyboard and printer wire pairs to their respective terminals and install page roll.

f. At TH-22( )/TG Positions 1, 2 & 3 install send/REC plugs and 4WS and 2W-4WR wire pairs to their respective terminals and jacks.

g. At TH-22( )/TG Position 1 place the following controls and make adjustments.

(1) Norm-Rec-Send Switch to Norm

(2) 4W-2W TEL switch to 4W

(3) Mode one-Mode two Switch to Mode one

(4) VF-20 Switch to 20

(5) On-Off Switch to ON

(6) Plug a Milliameter into Position 1, TH-22/TG, DCMON. Jack on patch panel.

(7) Obtain access to Int. bat loop current (1R12) adjustment on the bottom of the TD-22( )/TG Position 1 and adjust the control for a Multimeter indication of 21 milliamperes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

(8) Repeat steps 1 thru 7 above for TH-22/TG Positions 2 & 3.

(9) Place switches S1 & S2 on signal entrance box to CONNECTOR POSITION.

h. At CV-425 Position 1 to 6 install 2W 4WS and 2W-4WR wire pairs to their respective terminals.

i. At CV-425( )/V place the following controls:

(1) TP-TG switch to TP

(2) 2W-4W switch to 2W

(3) Sensitivity switch to Low

j. At SB-22( )/PT install four batteries BA-30 in battery case.

k. At night alarm battery box install two batteries BA-30.

4.8.2 SB-22( )/PT Switchboard, circuit operation.-

a. At Distribution Box J-1077/U strap the following binding posts:

(1) Line 1 to line 21

(2) Line 2 to line 22

(3) Line 3 to line 23

(4) Line 4 to line 24

b. At patch panel, using a patch cord, patch spare TH-22/TG line 4WS to spare line 21.

c. Depress the ring switch on the spare TH-22( )/TG.

d. Line 1. Signal indicator shall turn from black to white and buzzer shall sound at SB-22( )/PT Switchboard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

e. Insert Operators Plug in line 1 calling jack, signal indicator shall turn from white to black, and buzzer shall stop sounding.

f. Repeat b, c, d and e above for lines 2, 3 and 4 by patching at patch panel and ringing SB-22( )/PT switchboard as follows:

Patch Cord to	SB-22( )/PT Lines
Line 22	Line 2
Line 23	Line 3
Line 24	Line 4

g. At Distribution Box J-1077/U strap the following binding posts:

(1) Line 5 to line 21

(2) Line 6 to line 22

(3) Line 7 to line 23

(4) Line 8 to line 24

h. Repeat b, c, d and e above by patching the patch panel and ringing SB-22( )/PT Switchboard as follows:

Patch Cord to	SB-22( )/PT Lines
Line 21	5
Line 22	6
Line 23	7
Line 24	8

i. At Distribution Box J-1077/U strap the following binding posts:

(1) Line 9 to line 21

(2) Line 10 to line 22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

j. Repeat b, c, d and e above by patching the patch panel and ringing SB-22( )/PT Switchboard as follows:

Patch Cord to	SB-22( )/PT Lines
Line 21	9
Line 22	10

k. Remove all patch cords from patch panel and strapping from Distribution Box J-1077/U.

4.8.3 CV-425/U Converter-Telegraph Telephone Signal circuit operation.-

a. At Distribution Box J-1077/U strap the following binding posts:

(1) Line 11 to line 12

(2) Line 13 to line 14

(3) Line 15 to line 16

b. At patch panel, using a patch cord, patch spare TH-22( )/TG line 4WS to No. 1 CV-425 loop.

c. Depress the ring switch on the spare TH-22/TG.

d. Line 12 Signal indicator shall turn from black to white and buzzer shall sound at SB-22( )/PT Switchboard.

e. Insert operators plug in line 12 calling jack, signal indicator shall turn from white to black, and buzzer shall stop sounding.

f. Repeat b, c, d and e above for CV-425( )/U equipment 2 thru 6 by patching at patch panel and ringing SB-22( )/PT Switchboard as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

Patch Cord to	SB-22( )/PT Lines
No. 2 CV-425 Loop	11
No. 3 CV-425 Loop	14
No. 4 CV-425 Loop	13
No. 5 CV-425 Loop	16
No. 6 CV-425 Loop	15
SB-22 OPR	17

g. Remove all patch cords from patch panel and strapping from Distribution Box J-1077/U.

4.8.4 TT-76( )/GGC and TT-98( )/FG Teletypewriter circuit operations.-

a. At Distribution Box J-1077/U strap the following binding posts:

(1) Line 17 to line 20

(2) Line 18 to line 19

b. At TH-22/TG switch to 4W position.

c. Place the ON-OFF switch on the two teletypewriter switch boxes to NON-SECURE position.

d. Install a test tape in the transmitter distributor of TT-76( )/GGC position 1 and send.

e. Ten error free messages of the test tape shall be received simultaneously at position 2 of the TT-76( )/GGC tape and on TT-98( )/FG page copy Position 2.

f. Send a test message from the keyboard of the TT-76( )/GGC Position 1.

g. An error free test message shall be received by the TT-76( )/GGC tape and on TT-98( )/FG page copy position 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z.
Check the source to verify that this is the current version before use.

h. Repeat d, e, f and g above by sending from TT-76( )/ GGC position 2 and receiving at position 1.

i. Remove test tape and strapping from Distribution Box J-1077/U.

4.8.5 Telephone circuit operation.-

a. Connect a test telephone set TA-312( )/PT to line 25 on distribution box J-1077/U.

b. At intercom and telephone box place switch to phone position.

c. At telephone set TA-312( )/PT in shelter, lift handset and actuate ringer. Test telephone shall ring, established communications between telephones.

d. At intercom and telephone box place switch to OFF position. Red tel. indicator shall not light.

e. At test telephone, lift handset, actuate ringer. Buzzer shall sound, communications between telephones shall not be possible.

4.8.6 Intercommunication circuit operation.-

a. Connect a test intercommunication set LS-147( )/FI to line 26 on Distribution Box J-1077/U.

b. At intercom and telephone box place switch to INTERCOM position, red intercom indicator lamp shall light.

c. At intercom set in shelter, establish communications between intercoms.

d. At intercom and telephone box place switch to OFF position. Red intercom indicator shall not light.

e. At test intercom, communications between intercoms shall not be possible.

f. Disconnect all test equipment in distribution box J-1077/U and secure operational tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.9 Vehicular transportation test.- The shelter facility shall be secured (and blocked to prevent shifting) to a government furnished 1 1/2 ton, M-715, truck and transported over the specially prepared course described in 4.9.1 for 10 laps at each of the speeds 5 mph, 10 mph, 15 mph and 18-20 mph. A lap is defined as traversing the course in one direction. The facility shall meet the requirements of Paragraph 3.5 after completion of the test. (Note: The shelter facility shall be loaded to simulate the AN/TGC-30( ) using live and dummy loads (See 6.1 and 6.2b for this test).

4.9.1 Vehicular transportation course.- The course for the military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have 1 x 1 inch chamfers. The sixth and twelfth timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the vehicle passing over them and shall be anchored down securely by any effective means. The surface on which the timbers are placed shall be hard and level.

4.10 Rail transportation.- The shelter facility loaded as specified in Paragraph 4.9, secured to a flatcar (the test flatcar) and coupled to two other flatcars shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tiedown slings (contractor furnished) shall be as specified in Figure 5 and shall meet the provisions of the test as specified in Paragraph 4.10.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in Paragraph 4.10.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government technical representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars all with brakes off. All cars shall have standard (rigid) draft gear. The test shall consist of four impacts. Two impacts with the shelter facility positioned longitudinally on the flatcar, one from each end of the shelter facility and two impacts with the shelter facility positioned transversely on the flatcar, one from each side of the shelter facility. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of Paragraph 3.6.

4.10.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall be notched or shimmed to contact the trim along the sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred above. All other blocking shall be firmly secured as shown in the Figure 5. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter 6 x 19, flexible steel, preformed wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9000 pounds minimum. Before each impact, each sling shall be taut (600 - 800 pounds tension).

4.10.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

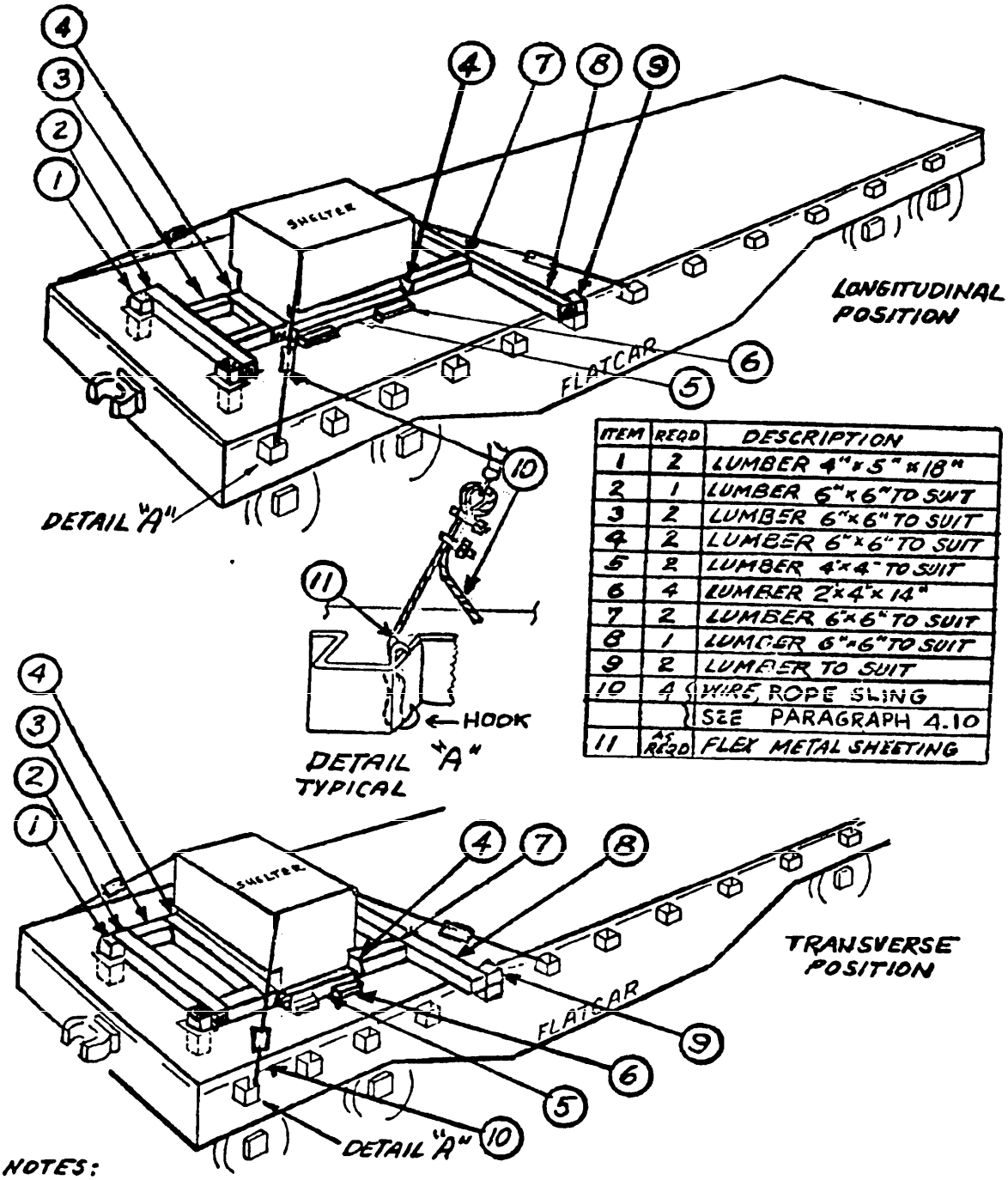
a. A coarse size of either coal, crushed stone, gravel, crushed concrete or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

4.11 Drop tests.-

4.11.1 Flat drop.- The shelter facility loaded as in 4.9 shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.



ITEM	REQD	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 6"x6" TO SUIT
5	2	LUMBER 4"x4" TO SUIT
6	4	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	4	WIRE ROPE SLING SEE PARAGRAPH 4.10
11	AS REQD	FLEX METAL SHEETING

NOTES:
1- THE BLOCKING LUMBER SHALL BE NOTCHED TO ACCOMODATE TOWING EYES, BOLTS AND RIVET HEADS.
2- THE BLOCKING LUMBER SHALL BE RIGIDLY FASTENED TO THE FLOOR OF THE FLATCAR.

FIG. 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

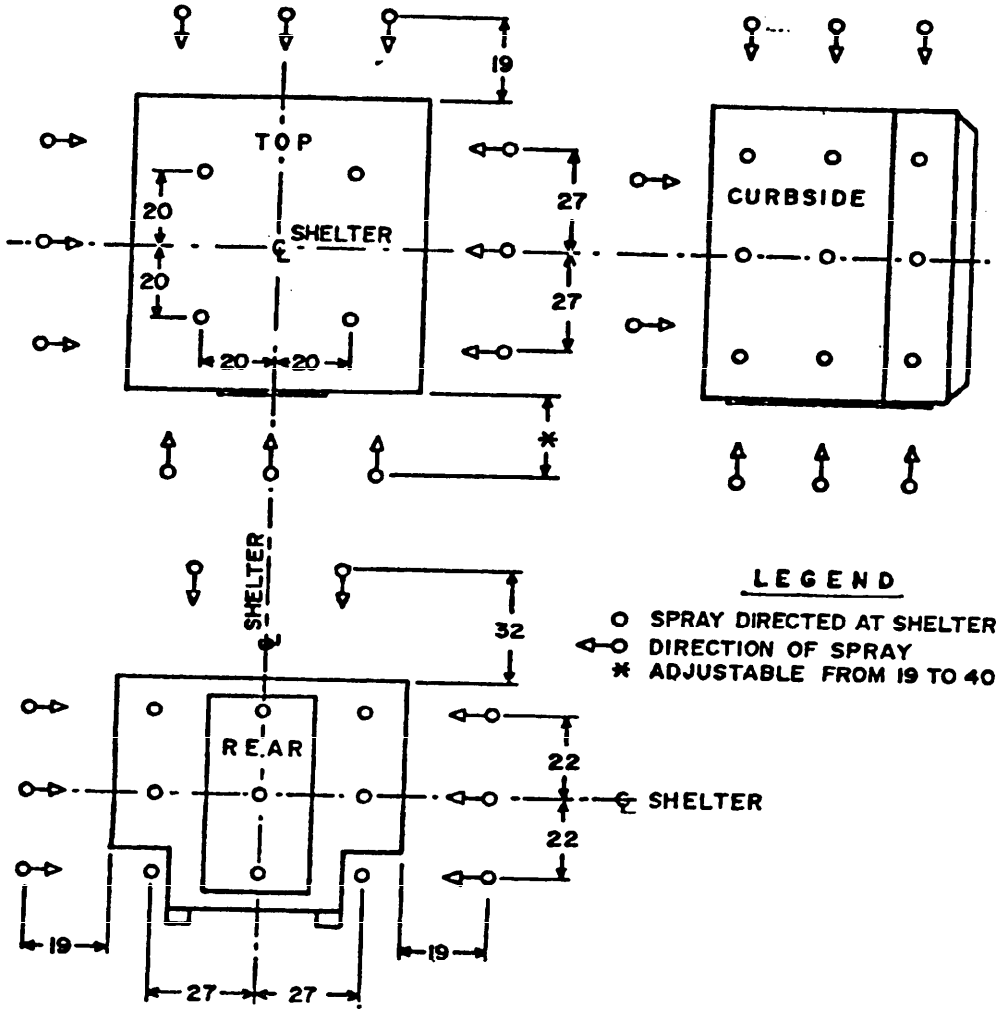
4.11.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid (or the knuckles of the skids) resting on a nominal 4 inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

4.12 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test (NOTE: Perform tests with inside covers of entrance boxes removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G 29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using nine nozzles, and the roof using four nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See Figure 6 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.13, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position (with exception of the exhaust blower vent covers at the front of the shelter and the door vent panel at the rear of the shelter) and repeat a above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



LEGEND

- O SPRAY DIRECTED AT SHELTER
- ⟵O DIRECTION OF SPRAY
- * ADJUSTABLE FROM 19 TO 40

NOTES

1. NOZZLES SHALL BE PERPENDICULAR TO SHELTER SURFACE.
2. UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3. GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4. ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5. FRONT NOZZLE LOCATIONS SAME AS REAR.
6. DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.13 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 26 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.14 Mounting insert tests.-

4.14.1 First Article and Group C insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

4.14.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any inset shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE V - Mounting Inserts Loads

Insert size	First Article and Group C Test		Production Test	
	Axial(lbs)	Torque(in.-lbs)	Axial(lbs)	Torque(in.lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.15 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.16 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks, patch panels and cabinets.

4.17 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VI.

TABLE VI - Classification of Visual & Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated.
	2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible.
	3. Welds - cracked or porous.
	4. Sealer improperly applied (holes, separations, of lack of adhesion).
	5. Mounting inserts missing or mislocated.
	6. Burrs or sharp edges in wire duct not removed.
	7. Equipments or bracketry loose or missing.
	8. Doors, etc. inoperative.
	9. Electrical receptacles, switches, connectors, wire ducts or other electrical components loose, improperly located or inoperative.
	10. Locking or holding devices missing or inoperative.
	11. Gaps between sections or wire duct exceed 1/16".
	12. Gaskets, missing.
Minor	1. Doors, not easily operable.
	2. Fastening devices difficult to operate.
	3. Finish - abrasions or scratches.
	4. Burrs and sharp edges on bracket not removed.
	5. Screws loose.
	6. Gaskets, not completely and securely adhered.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.18 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.19 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.7	Power & Signal wiring
Paragraph 4.17	Visual and mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with applicable requirements of Specification MIL-S-55505.

6. NOTES

6.1 Intended use.- The Shelter, Electrical Equipment S-391( )/TGC-30 is the shelter facility portion of the AN/TGC-30( ) and it is used to house the operational equipments including:

a. 2 ea TT-98( )/TG Teletypewriter

b. 2 ea TT-76( )/GGC Teletypewriter Reperforator Transmitter.

c. 1 ea SB-22( )/PT Switchboard Telephone Manual

d. 3 ea TH-22/TG Terminal Telegraph

e. 6 ea CV-425/U Converter Telegraph-Telephone Signal

f. 1 ea TA-312( )/PT Telephone

g. 1 ea LS-147( )/FI Intercom

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

h. 2 ea TSEC/KG-7( ) Security Equipment (optional) forming the Central Office, Teletypewriter

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Preproduction and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads for the TSEC/KG-7( ) Security equipment (2 ea) and government furnished live loads (converters, telephone, teletypewriters, air-conditioner and components) unless otherwise specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-391( )/TGC-30( ) on order.

d. A list of equipments to be government loaned for testing purposes.

e. Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination, testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-391F/TGC-30. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army-EL

Preparing Activity:
Army - EL

Proj Nr. 5410-A167

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.